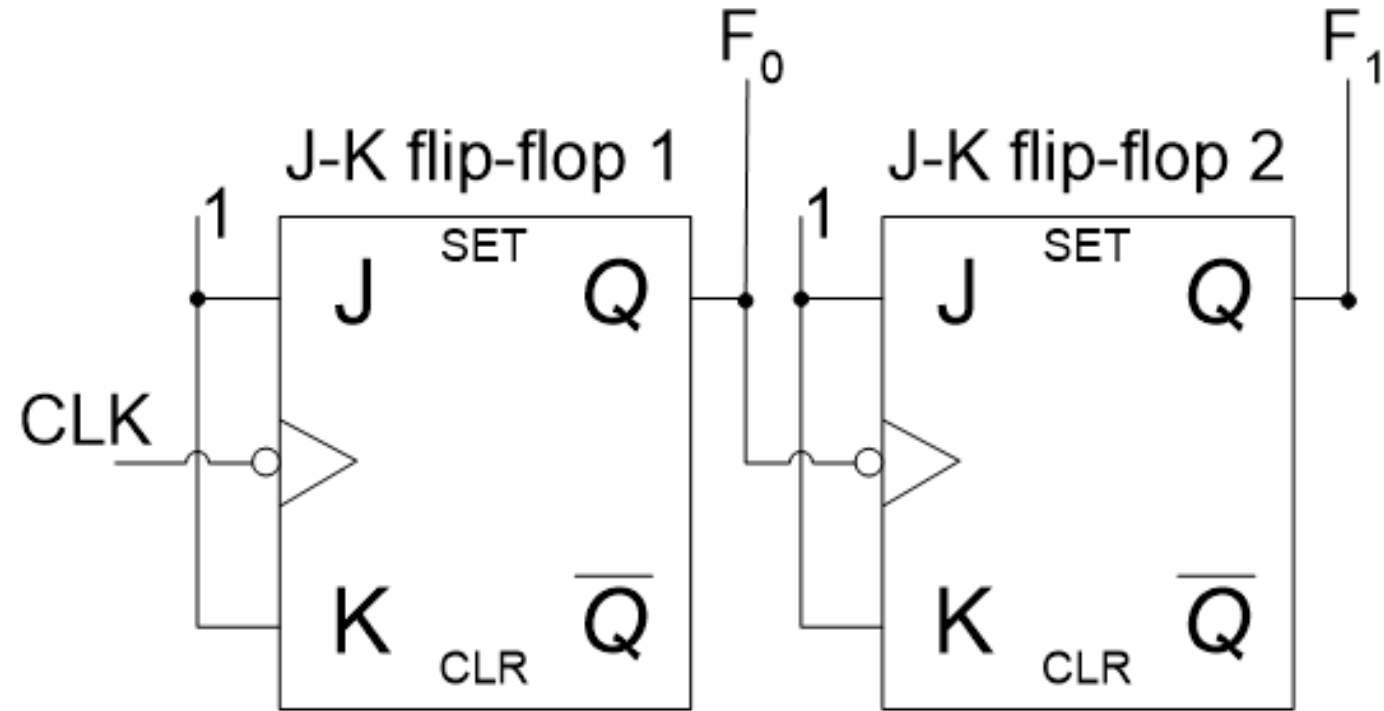


RIPPLE COUNTERS

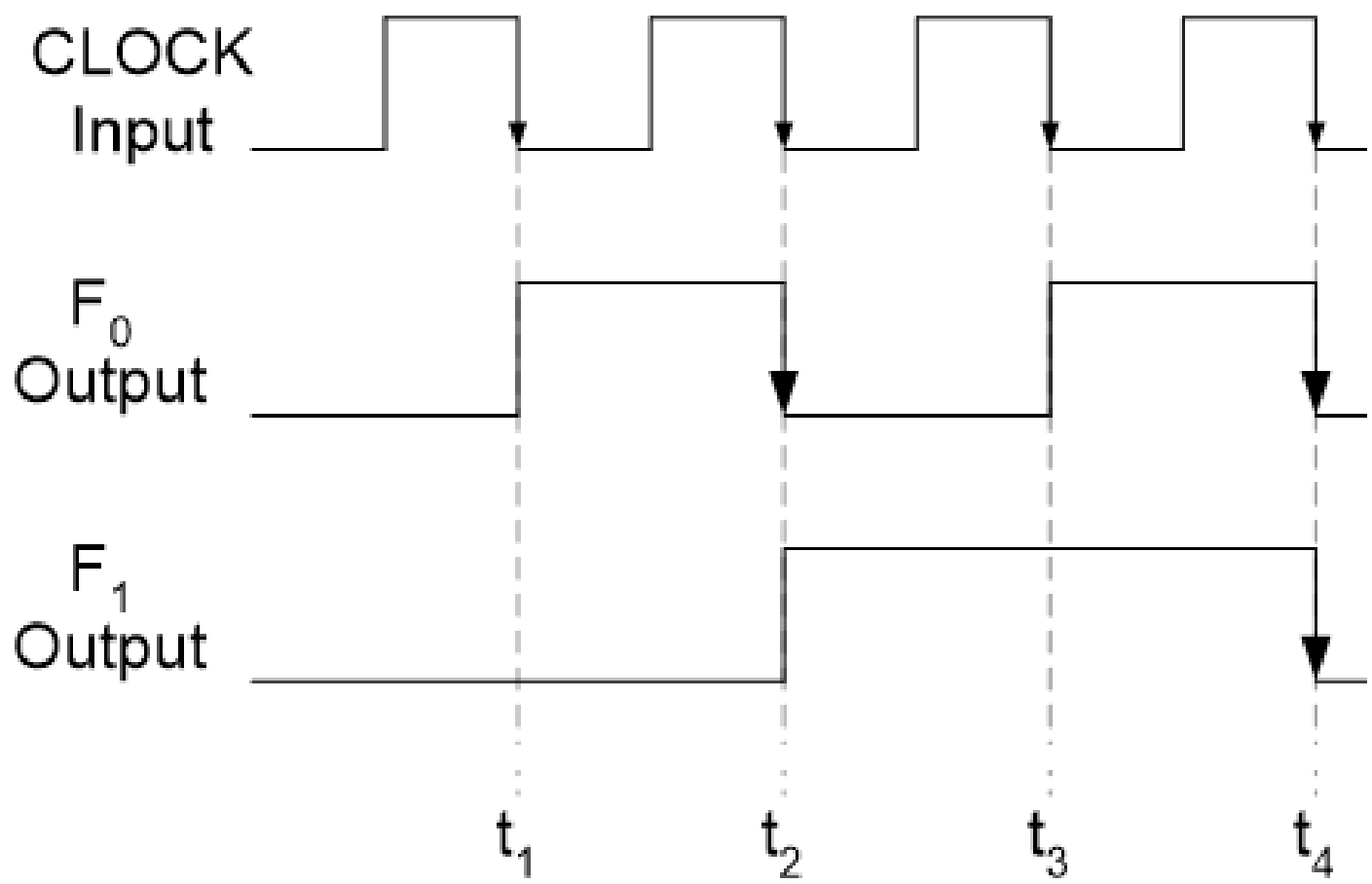
LECTURE 8

2-bit Asynchronous Up-Counter

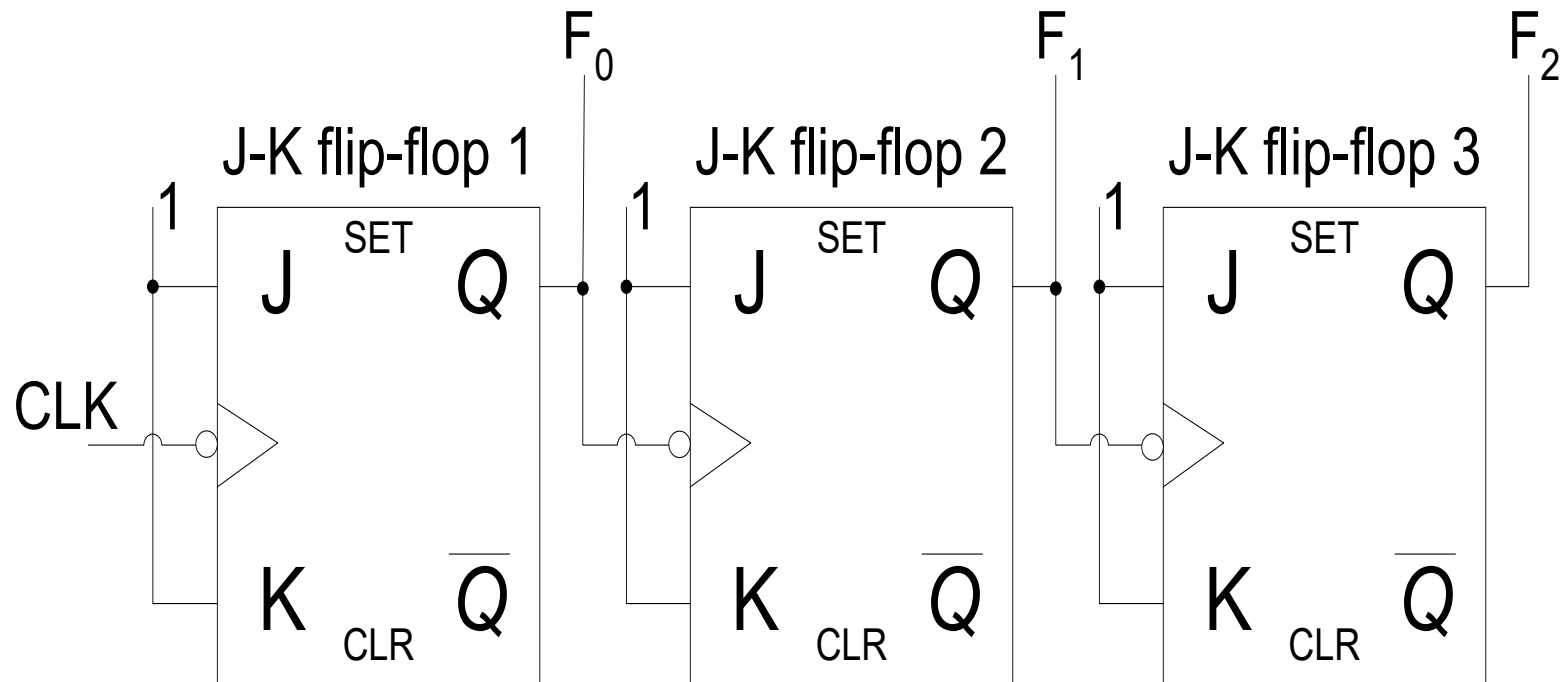


Timing Diagram

2-bit Asynchronous Up-Counter

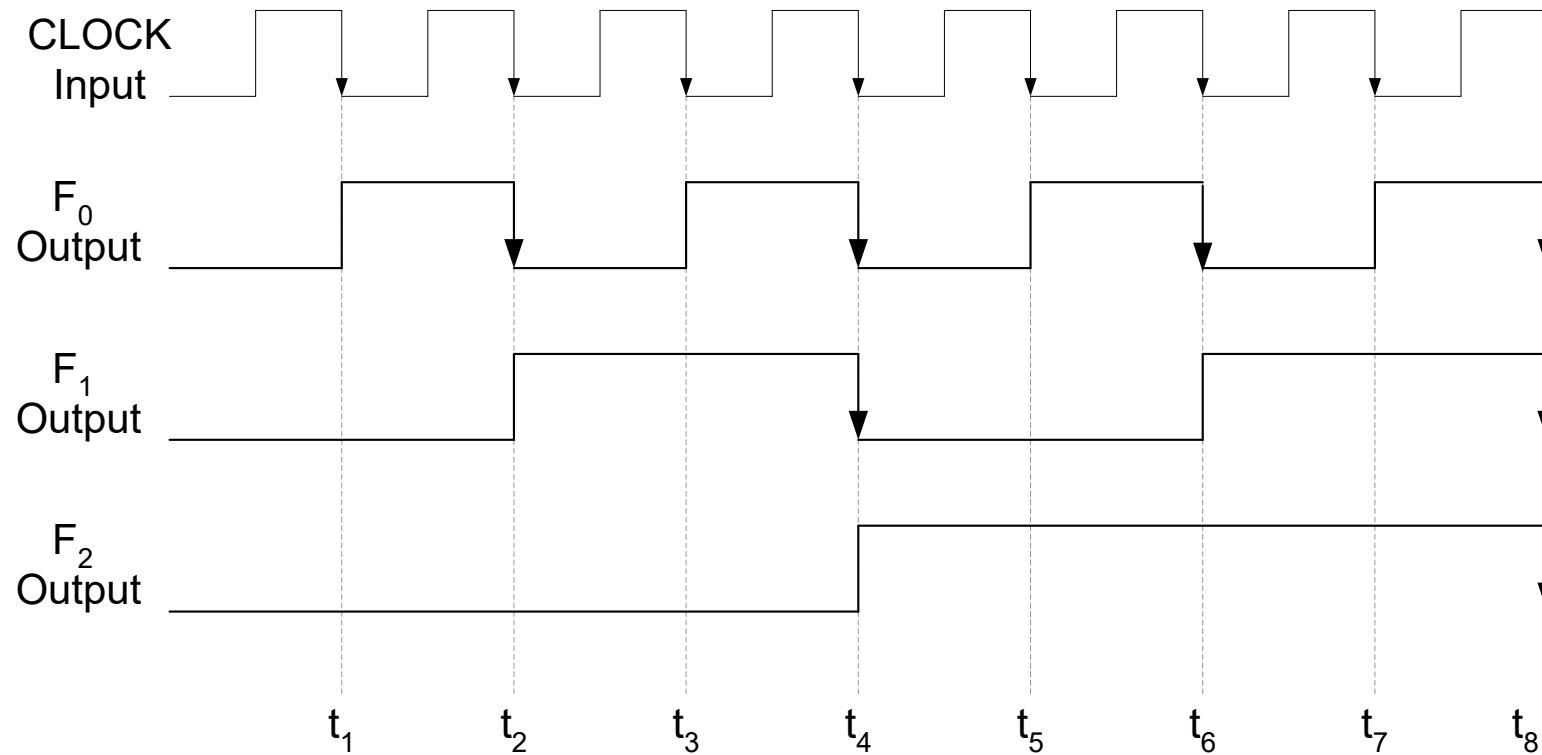


3-bit Asynchronous Up-Counter

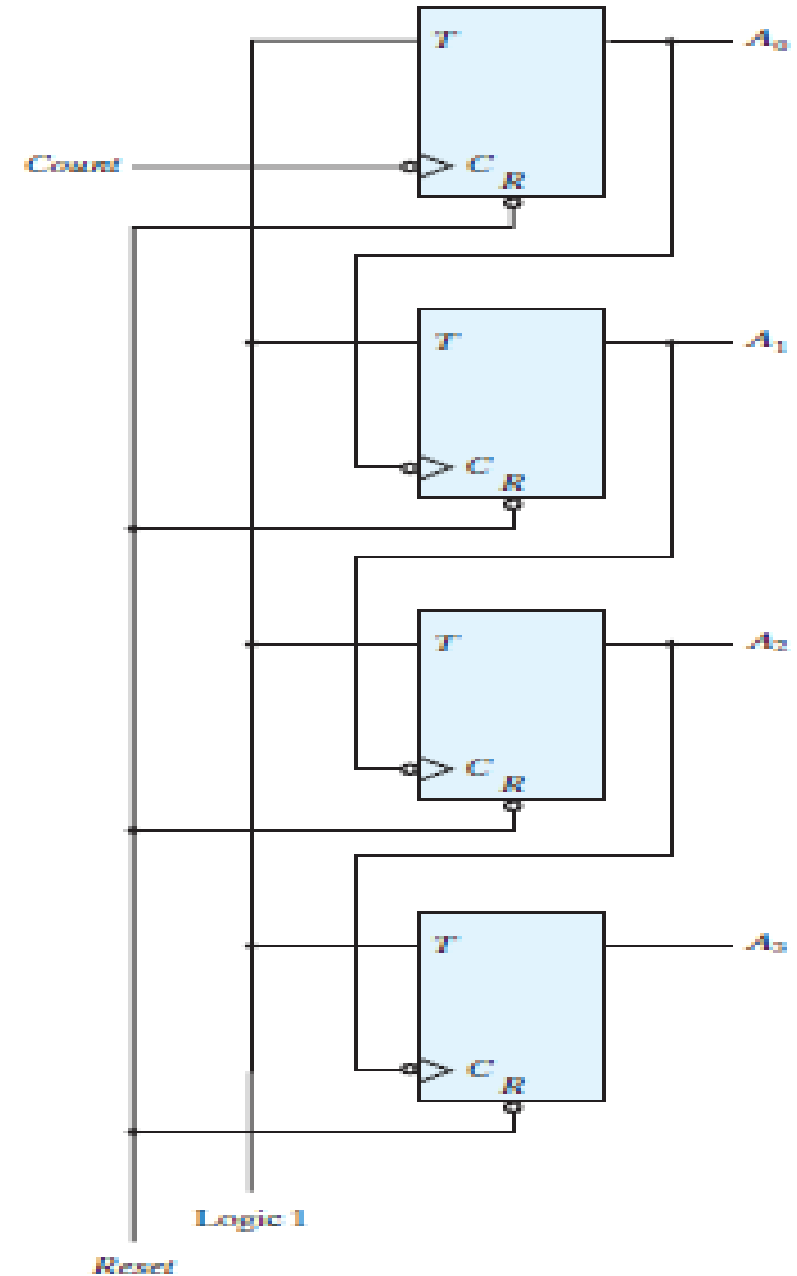


Timing Diagram

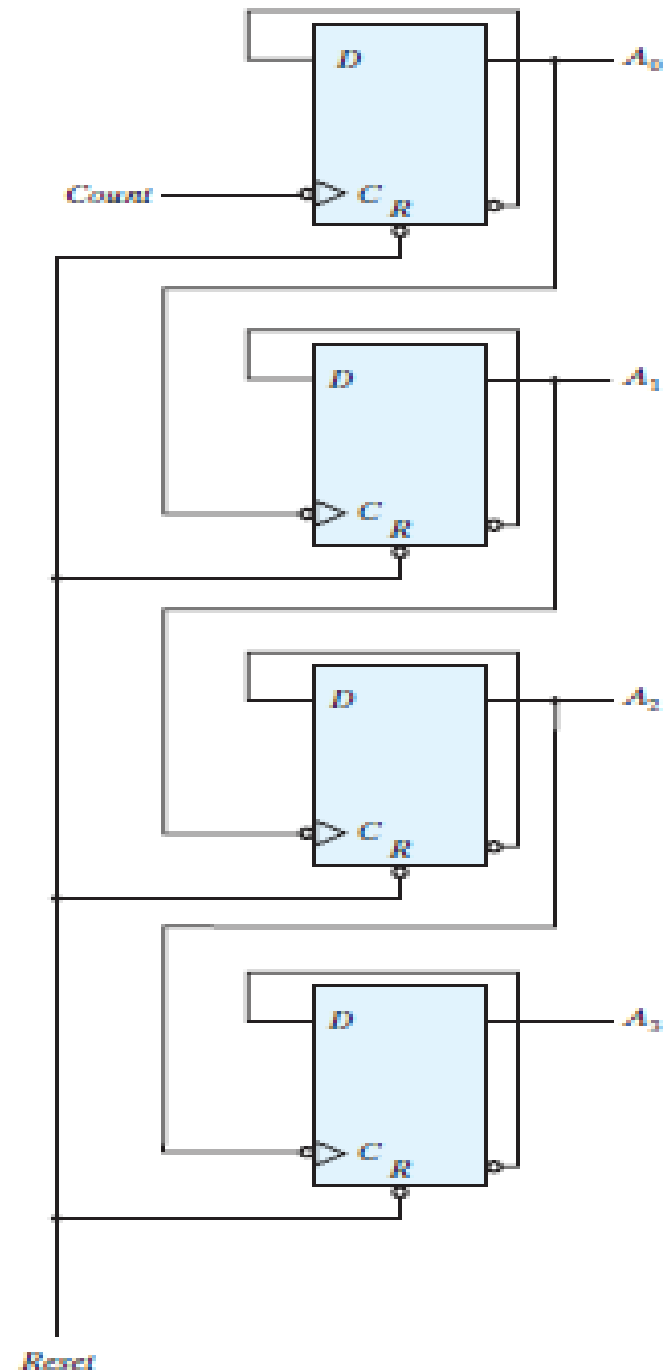
3-bit Asynchronous Up-Counter



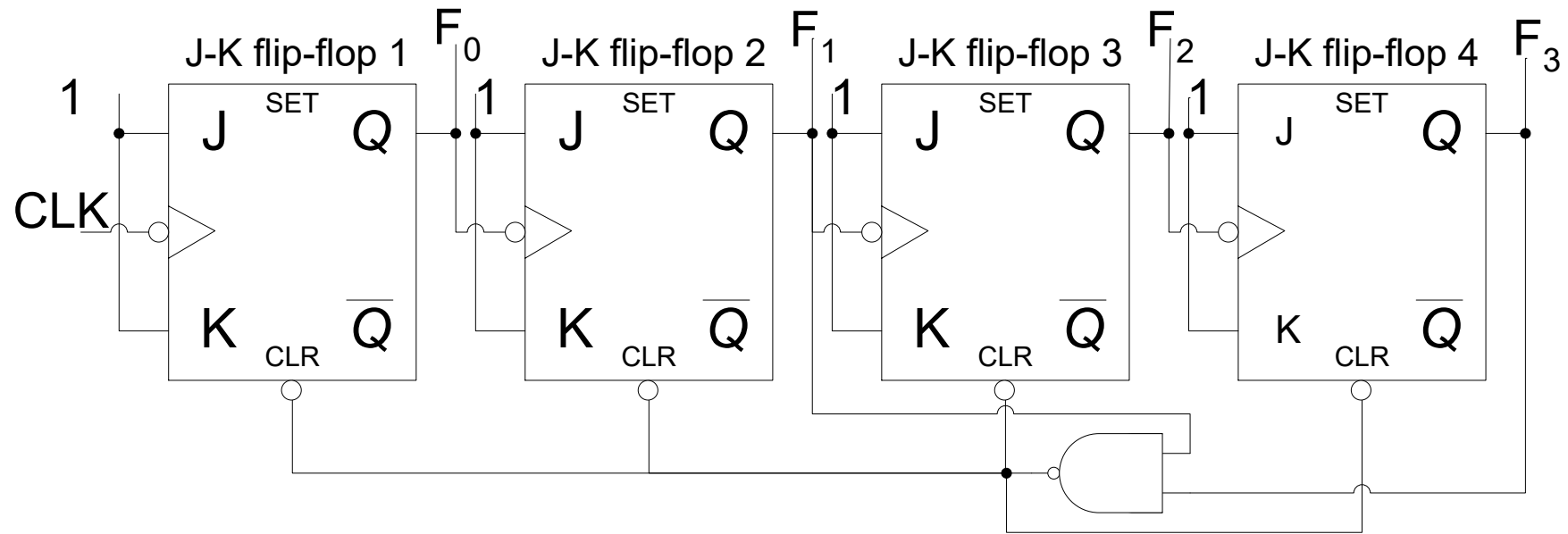
FOUR BIT BINARY RIPPLE COUNTER with T Flip Flop



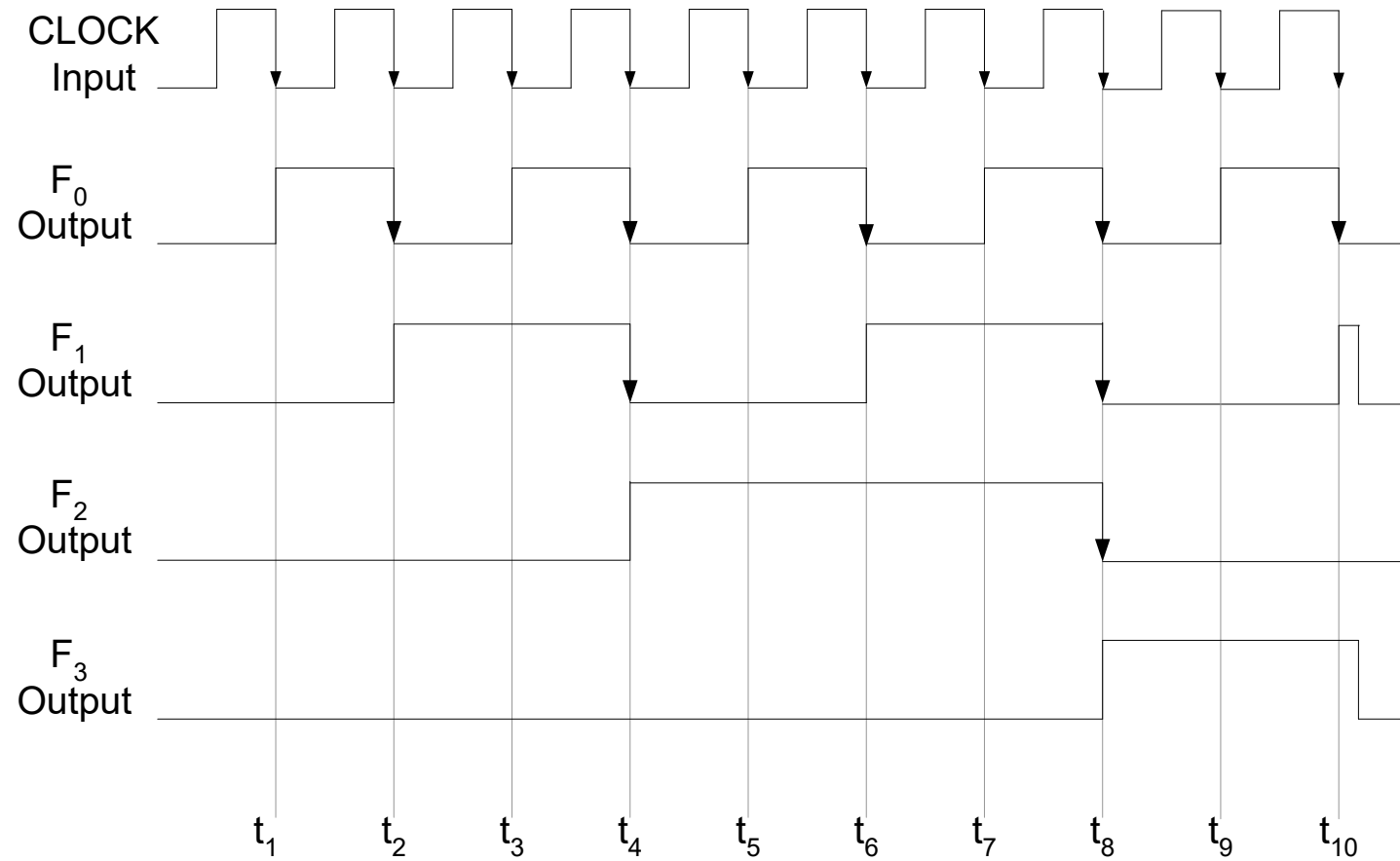
FOUR BIT BINARY RIPPLE COUNTER with D Flip Flop



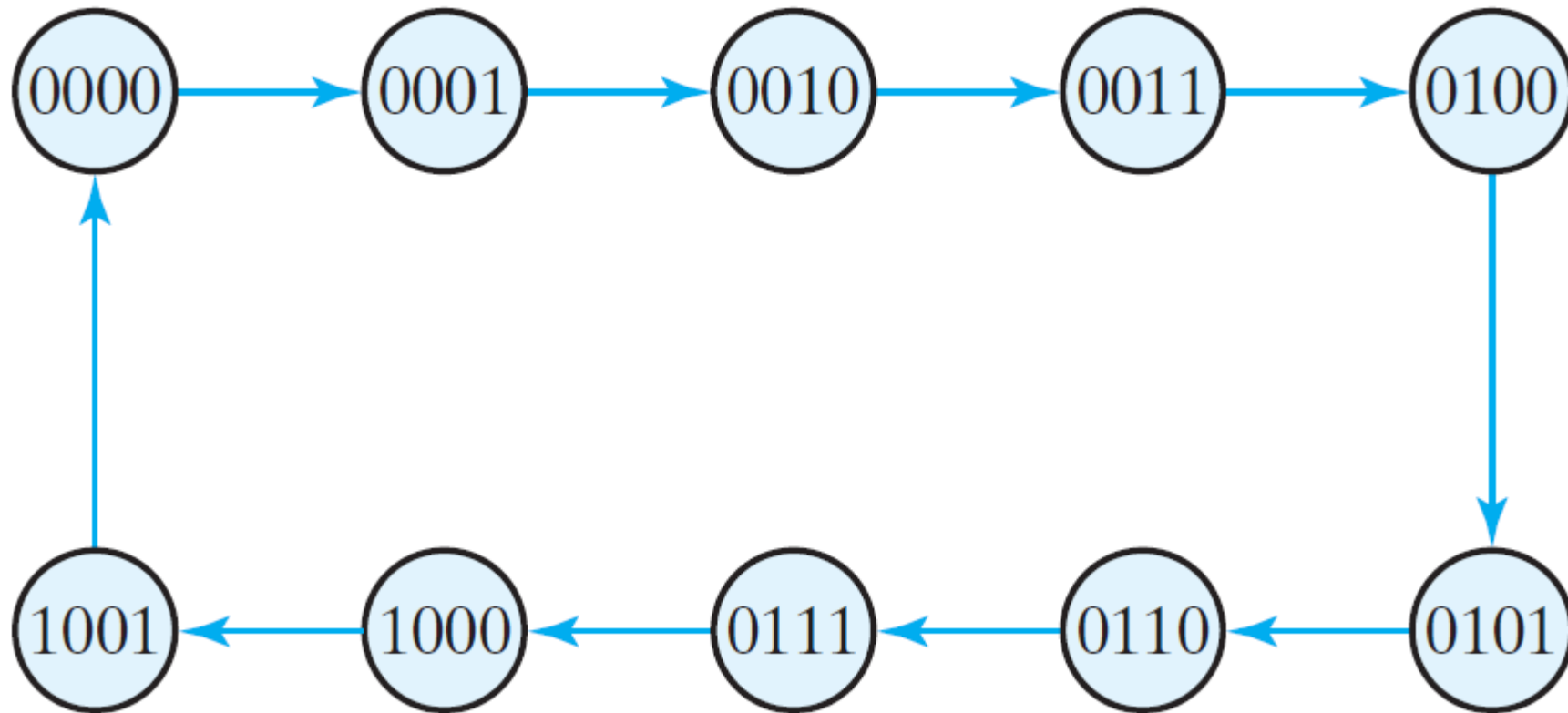
Asynchronous Decade Counter



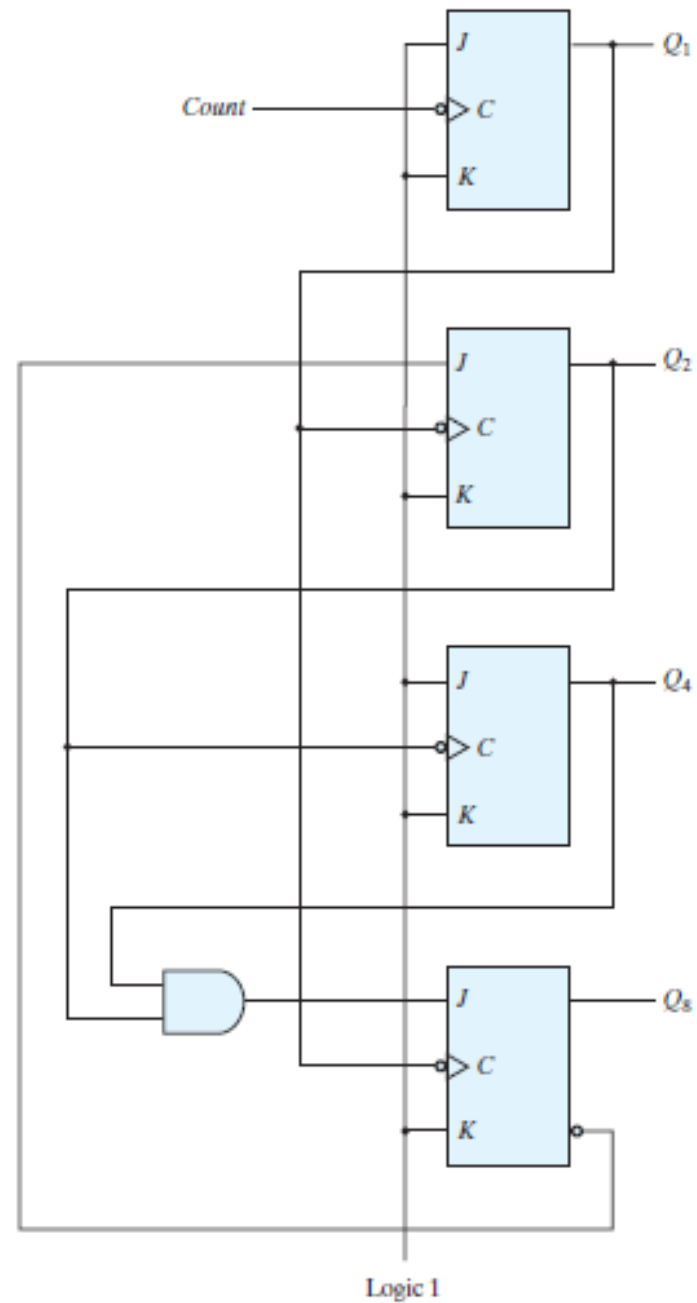
Timing diagram of a Decode Counter



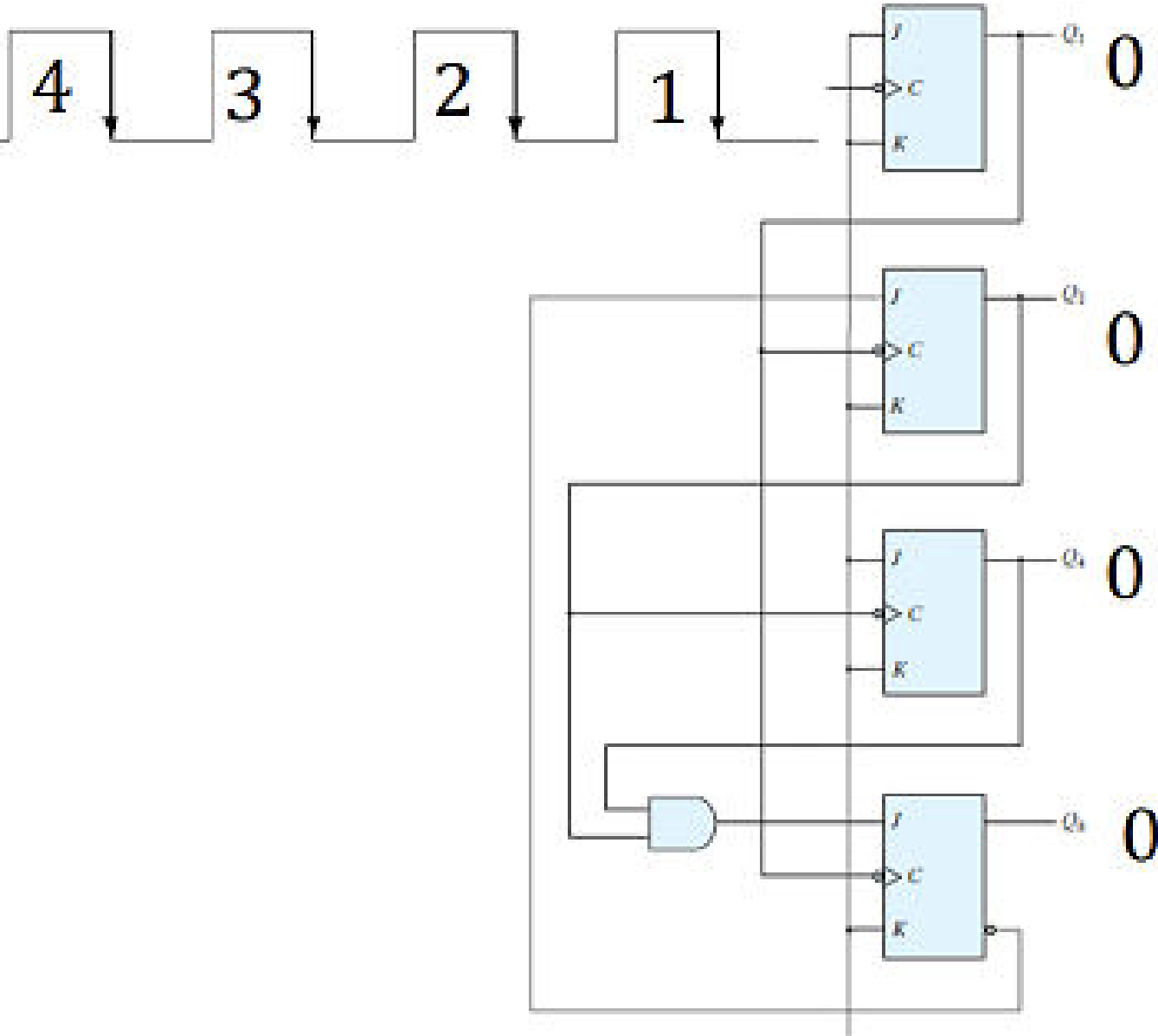
BCD Ripple Counter



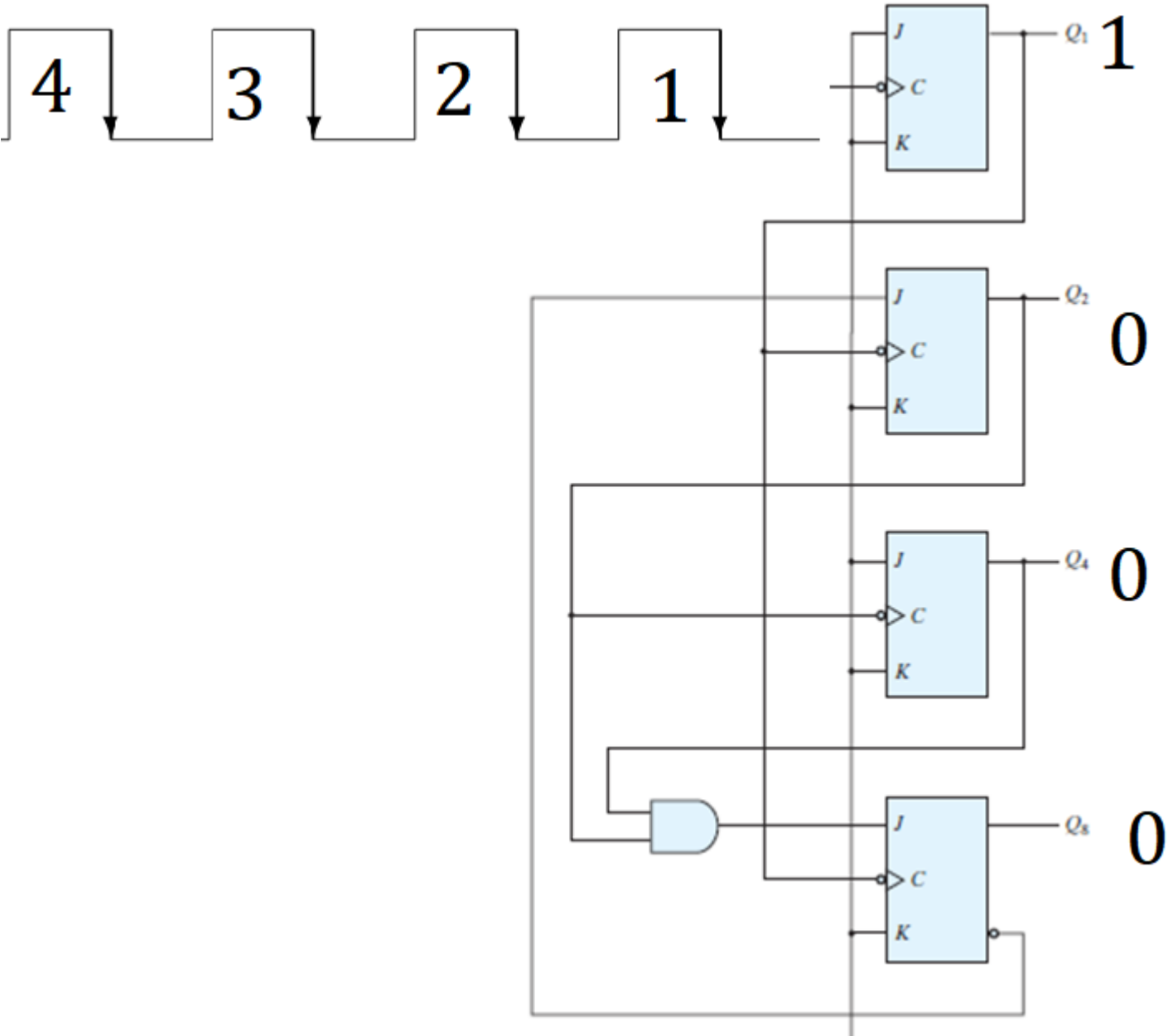
BCD Ripple Counter



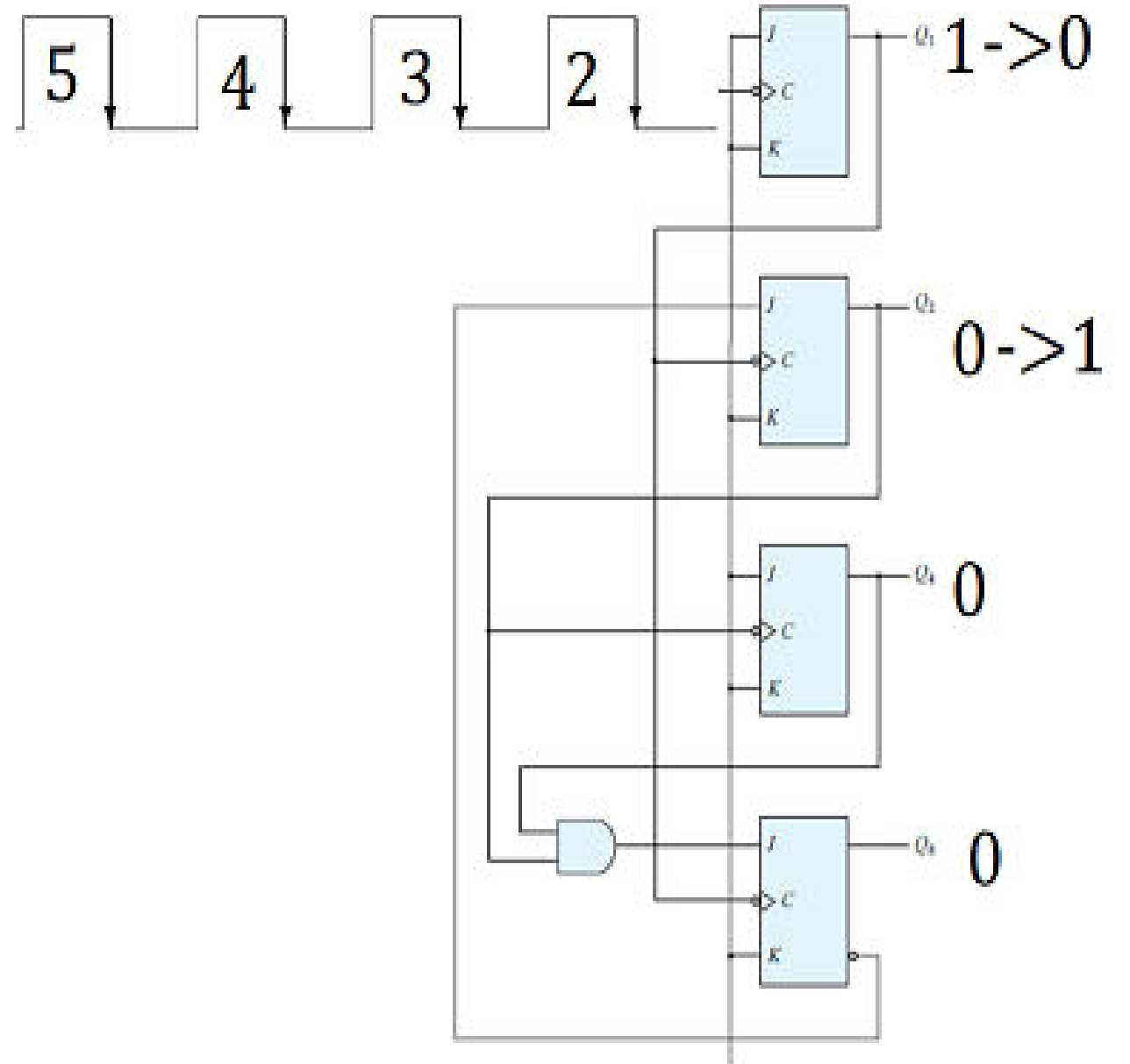
BCD Ripple Counter



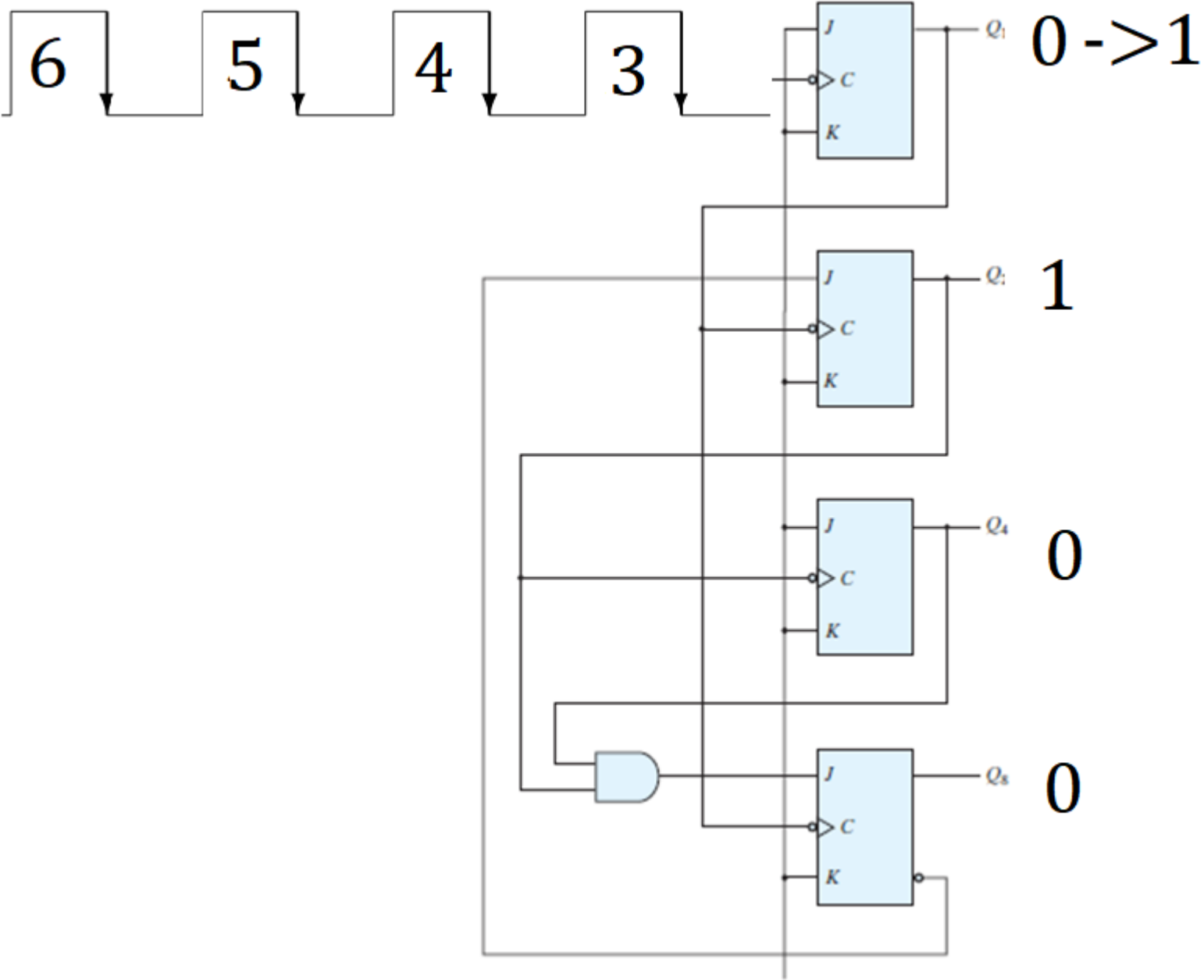
BCD Ripple Counter



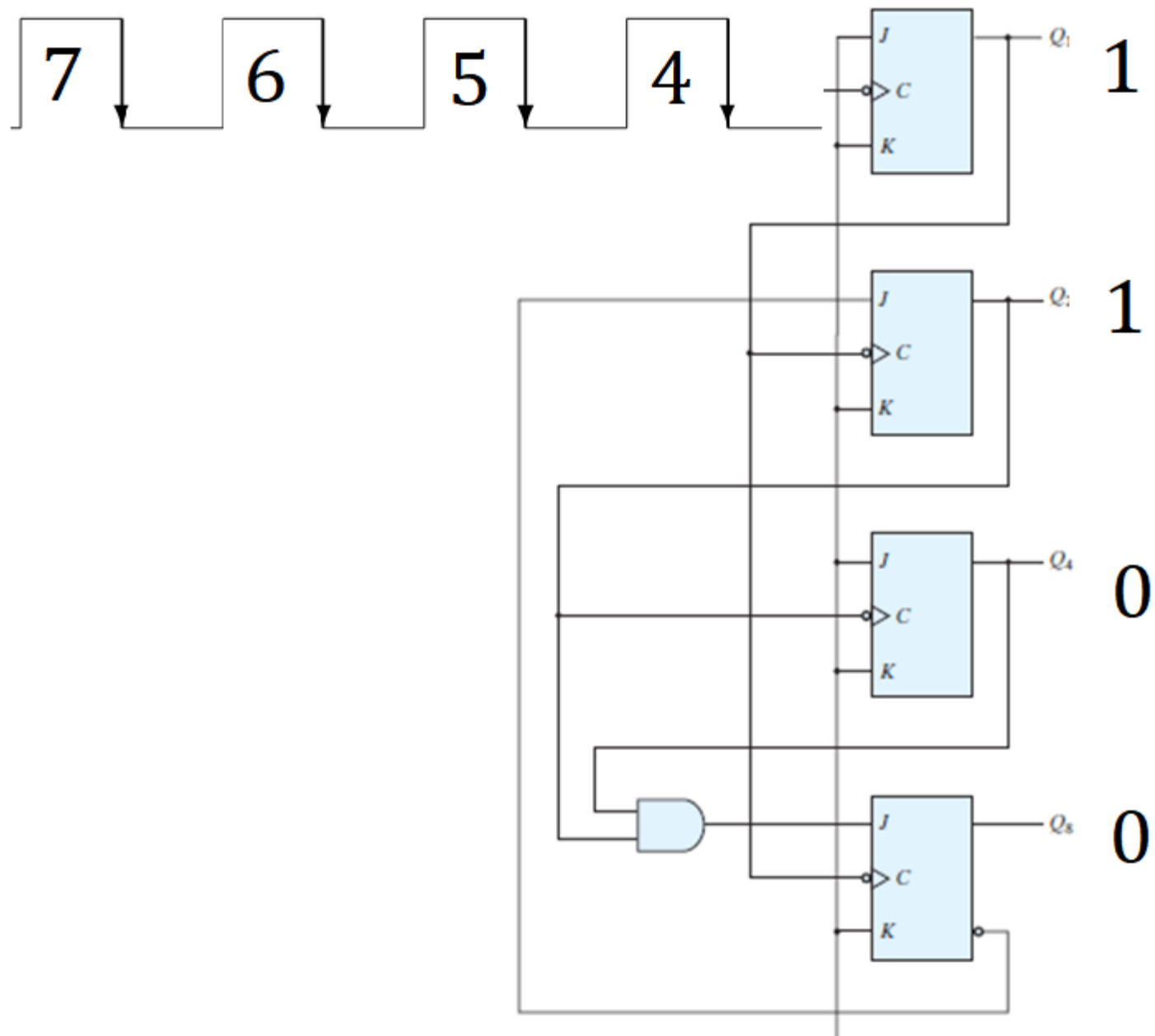
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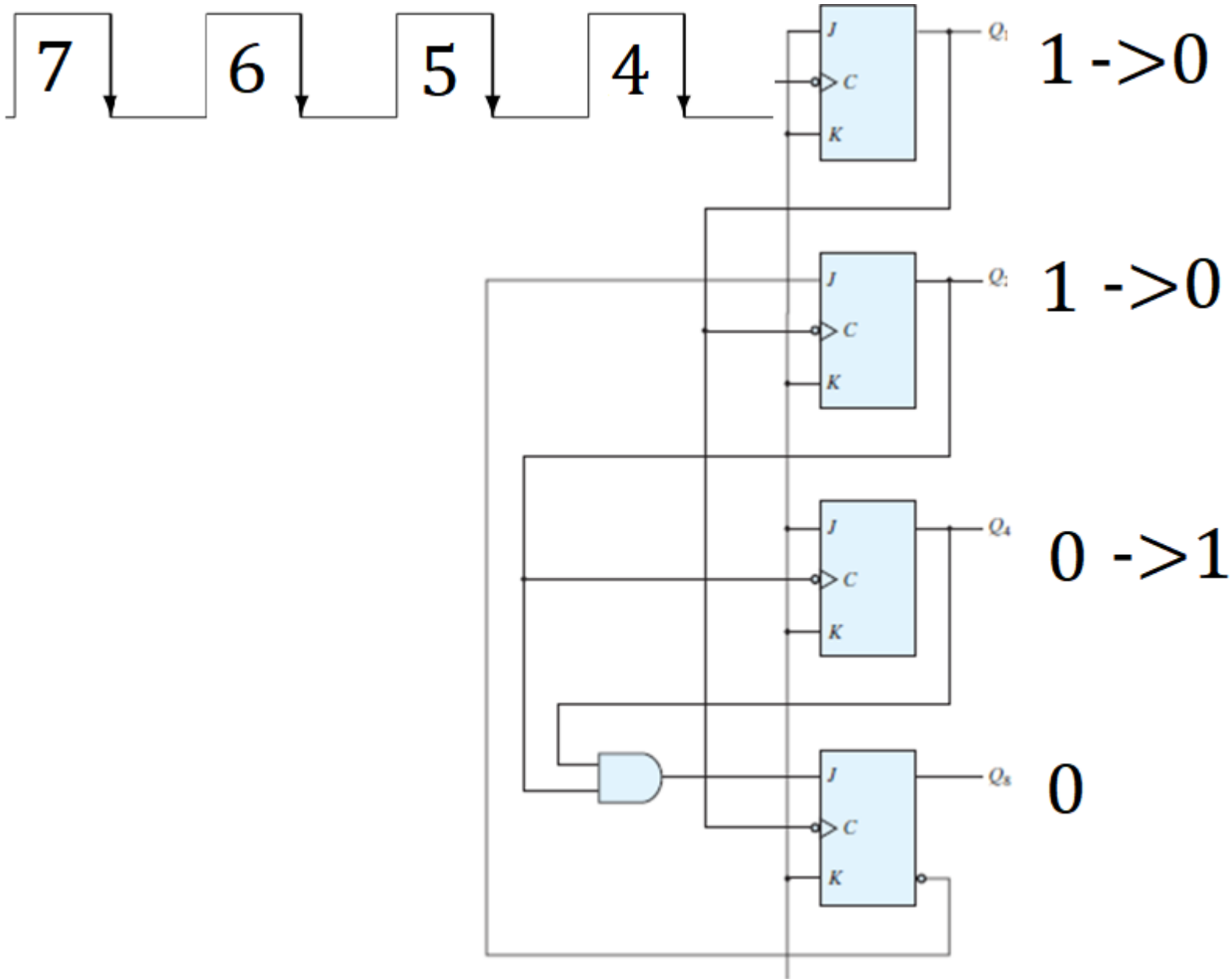
BCD Ripple Counter



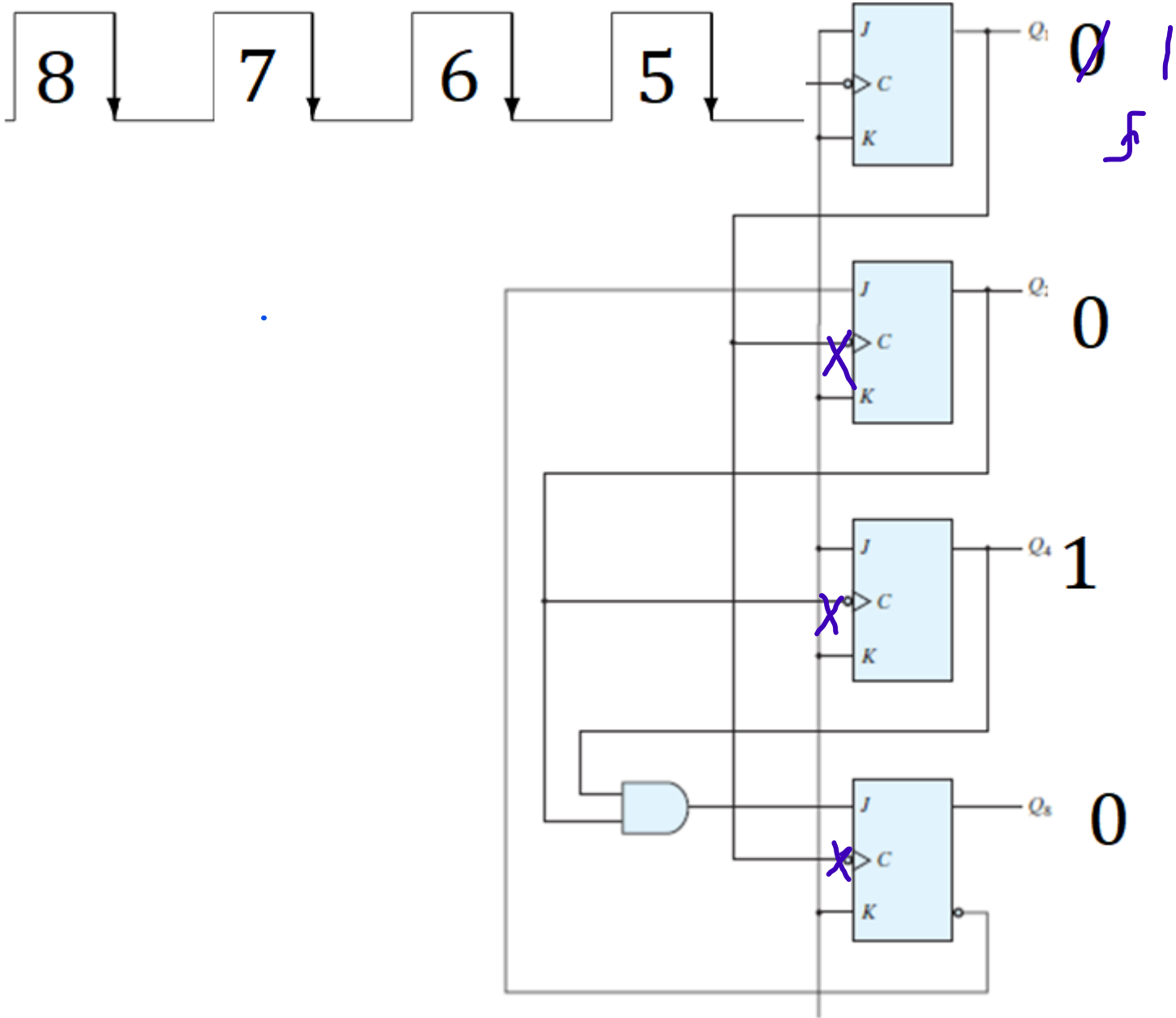
BCD Ripple Counter



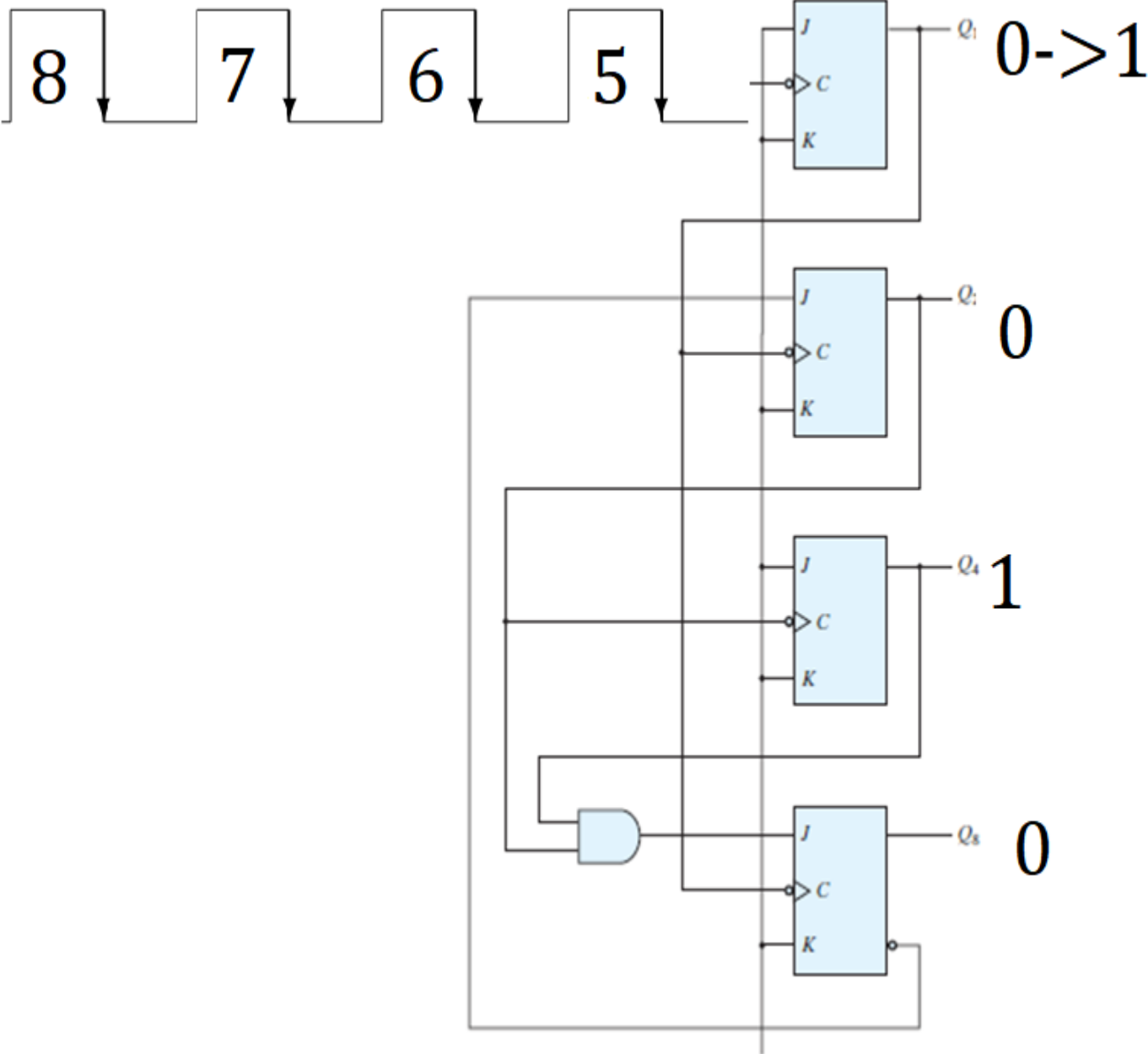
BCD Ripple Counter



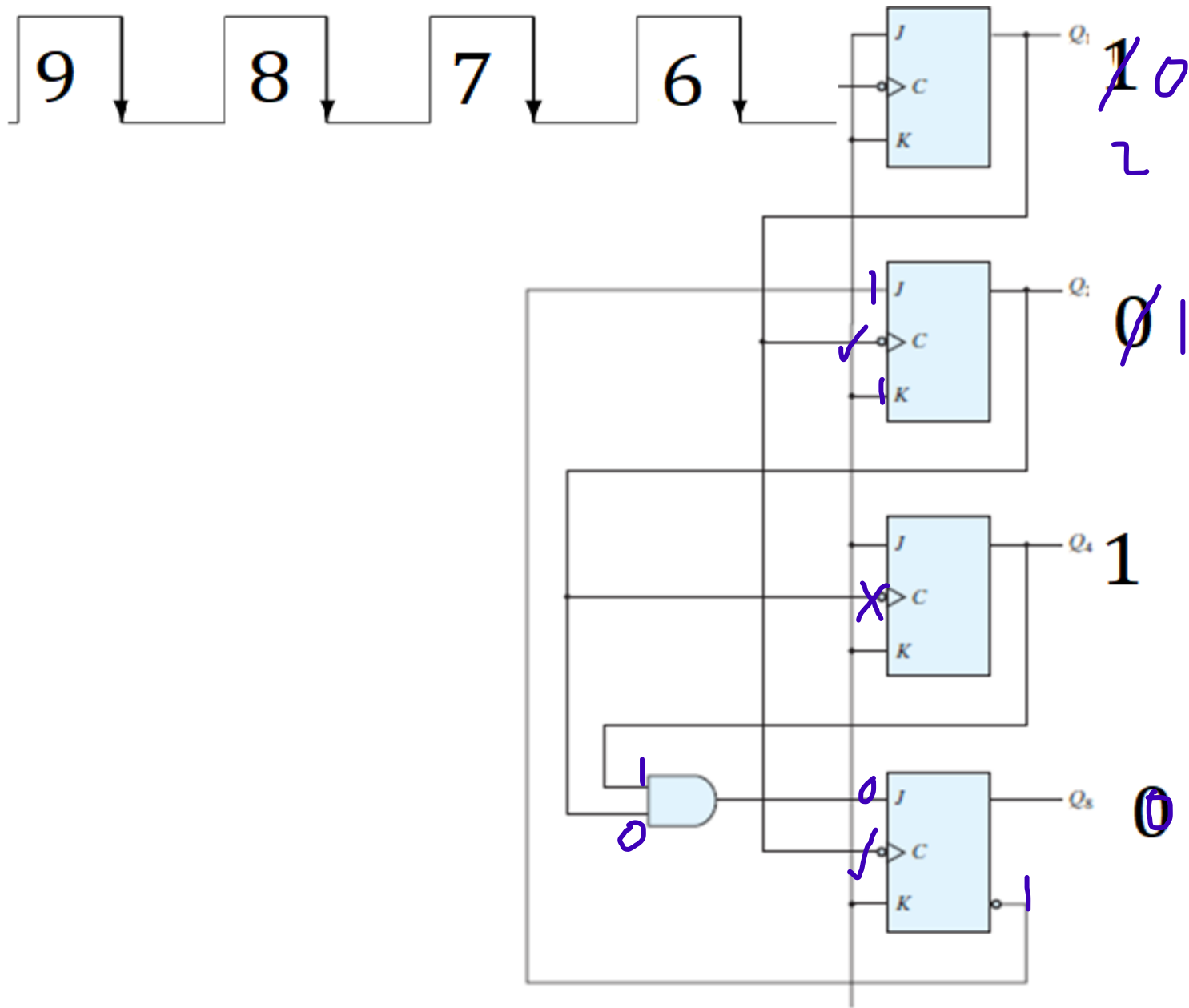
BCD Ripple Counter



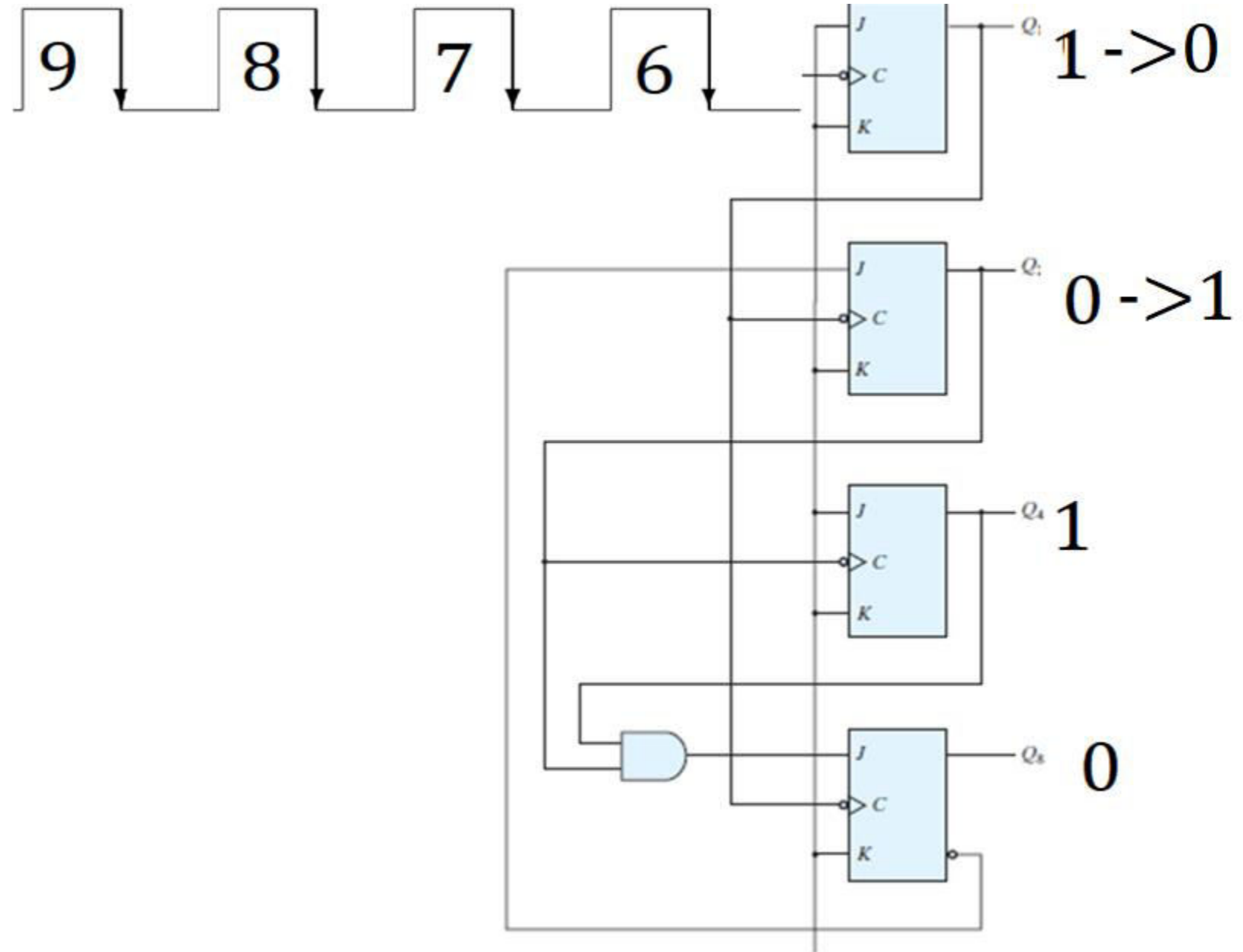
BCD Ripple Counter



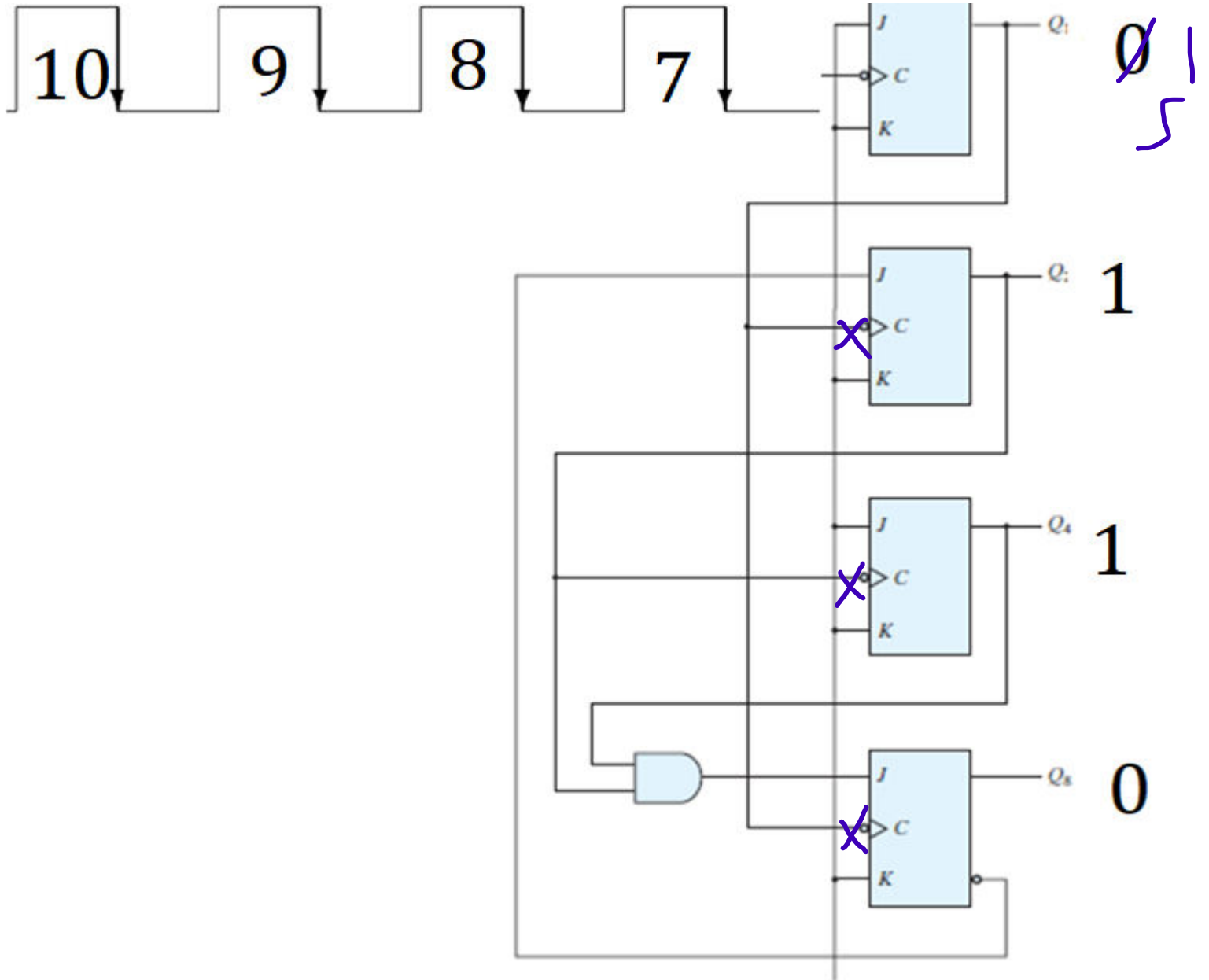
BCD Ripple Counter



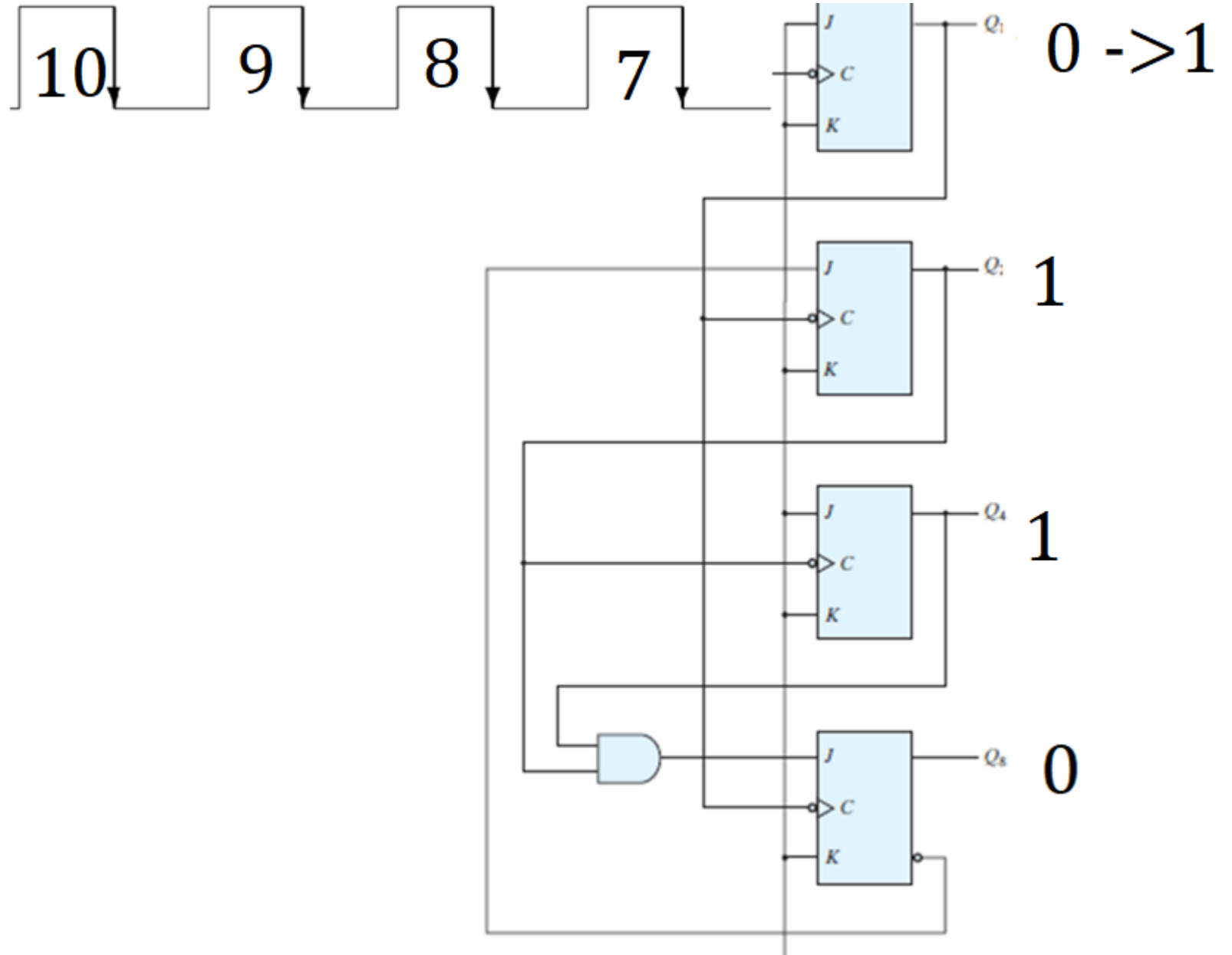
BCD Ripple Counter



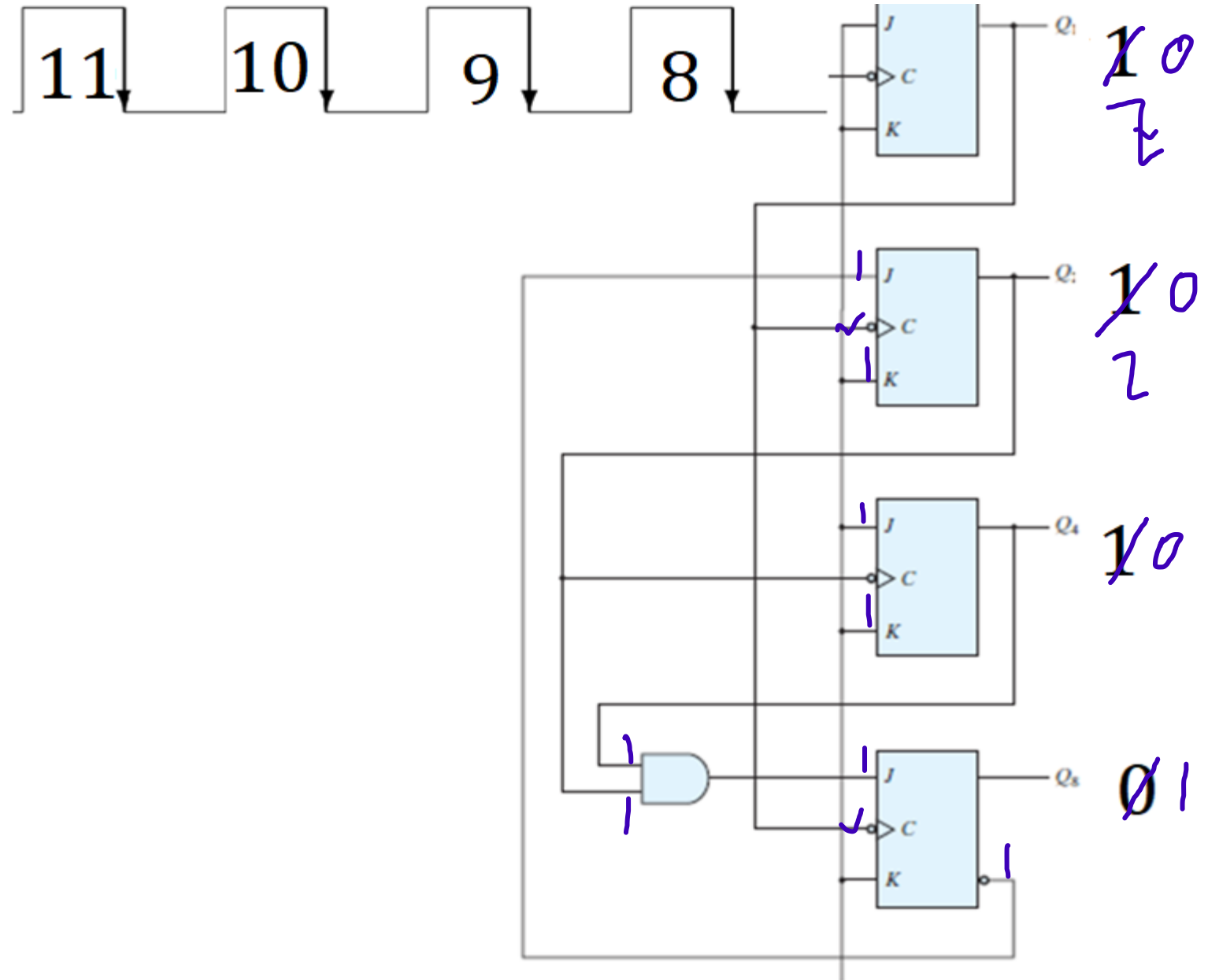
BCD Ripple Counter



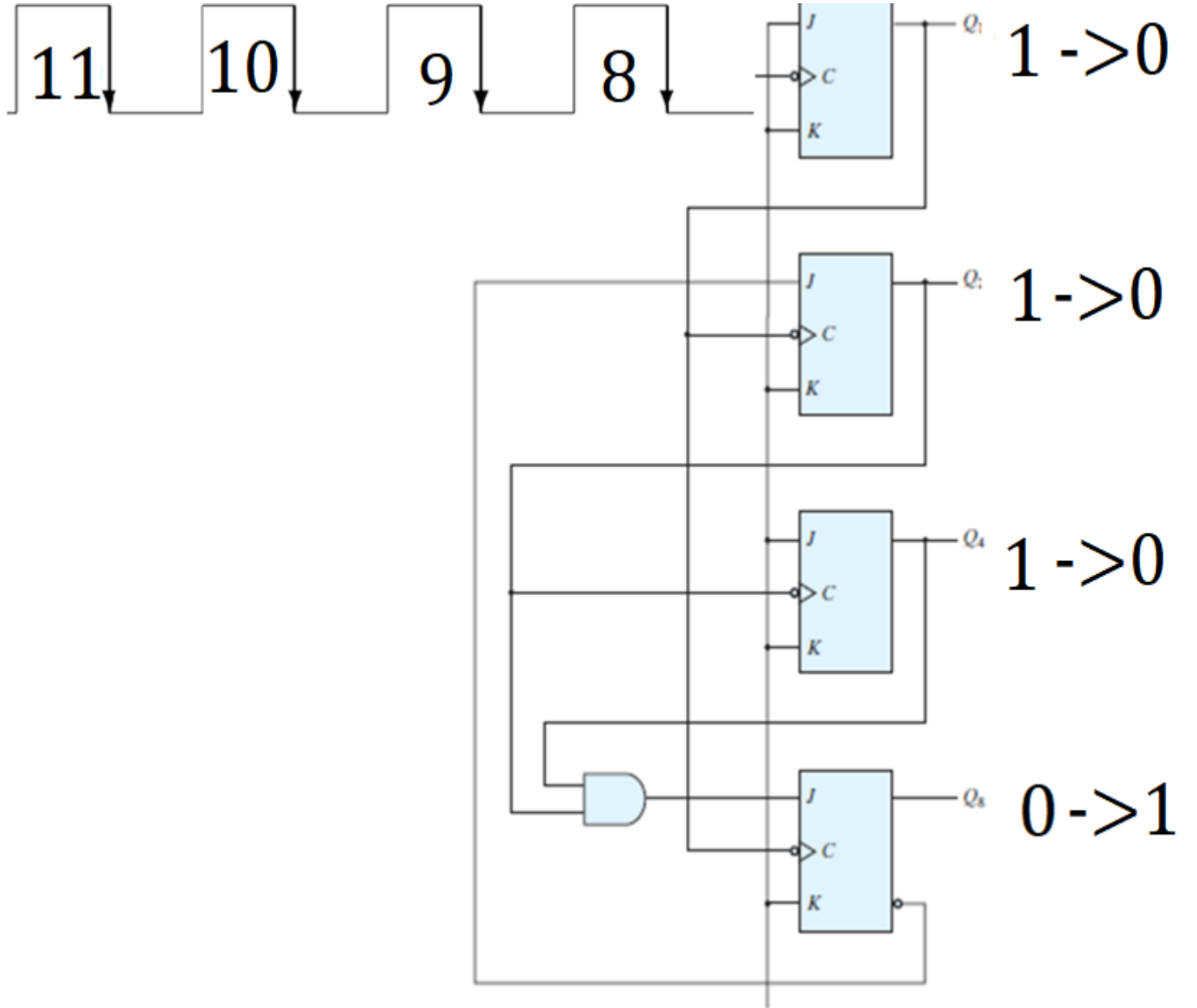
BCD Ripple Counter



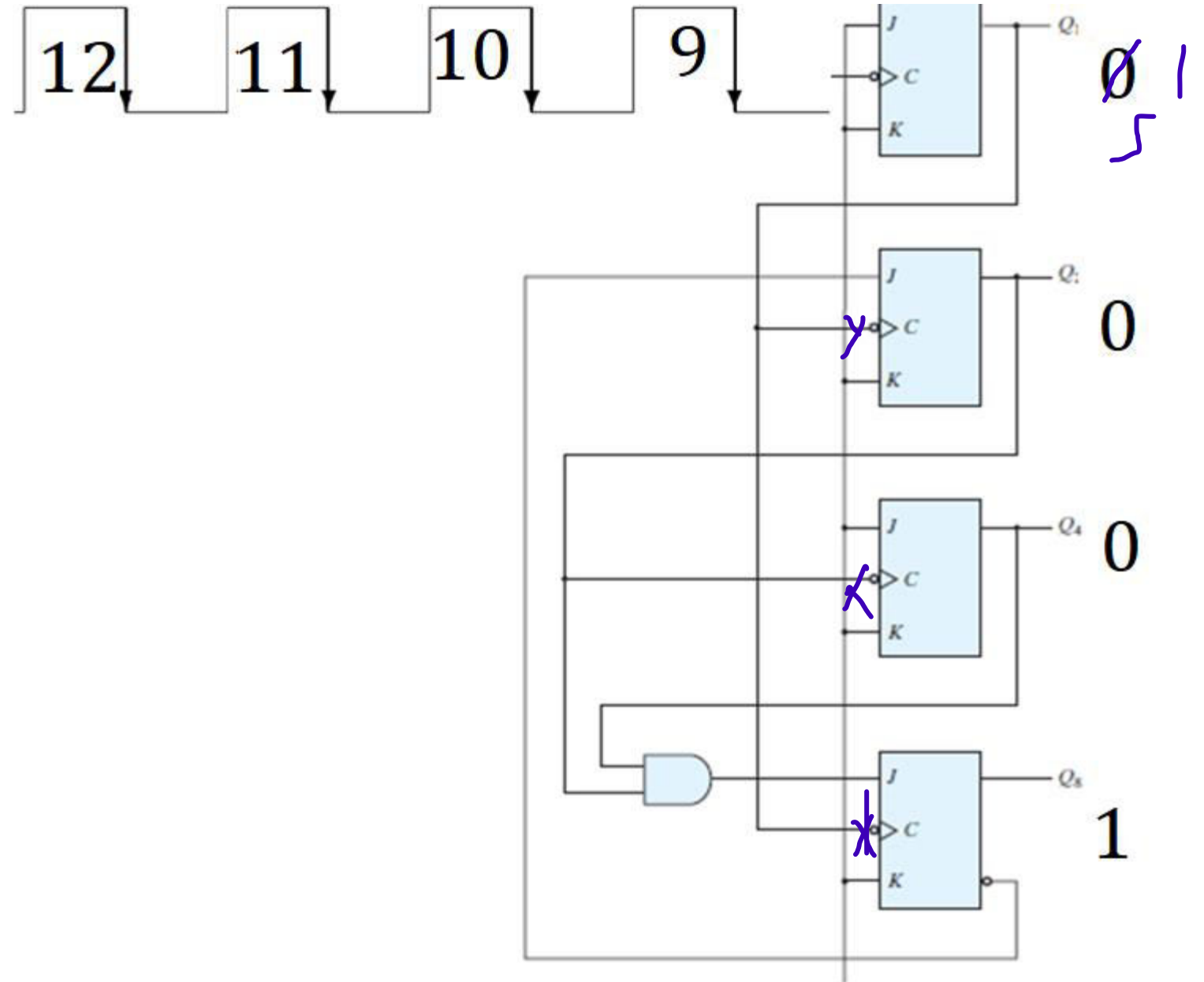
BCD Ripple Counter



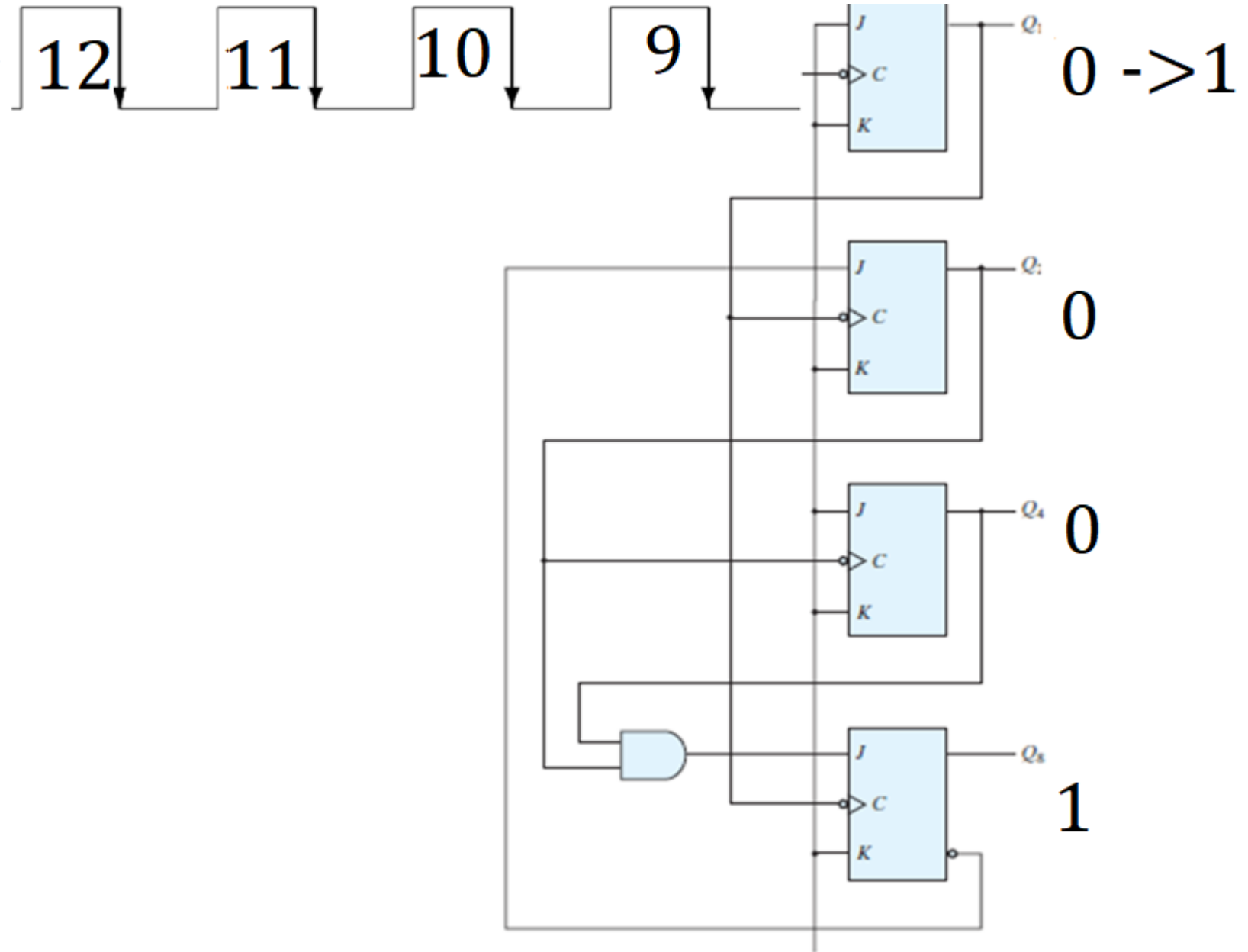
BCD Ripple Counter



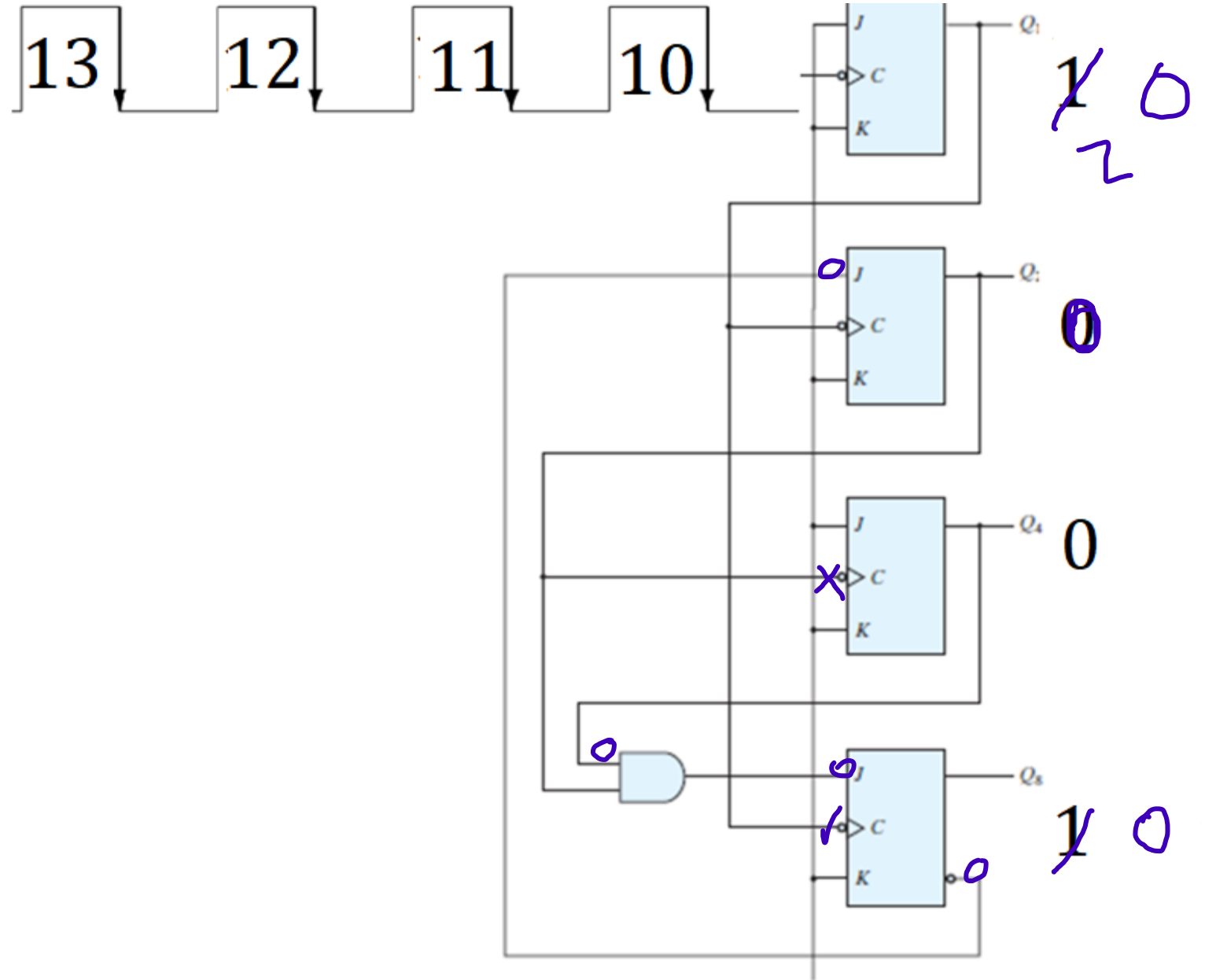
BCD Ripple Counter



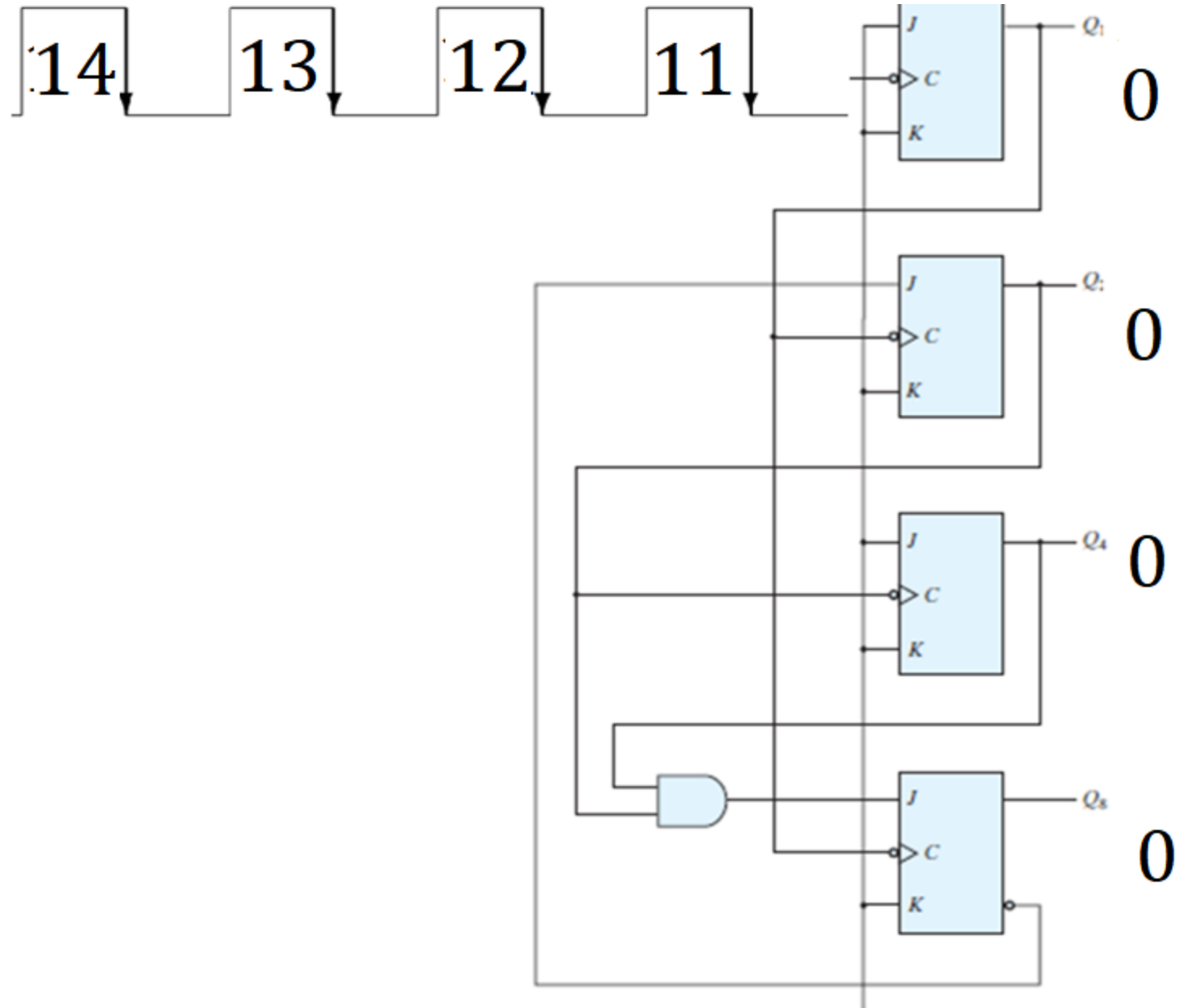
BCD Ripple Counter



BCD Ripple Counter



BCD Ripple Counter



THANKS

